

Breaking Information Silos in Smart Metro Security: An Edge–Cloud Collaborative AIoT Framework With Humanoid Agents

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Abstract—Conventional metro security screening systems rely on isolated X-ray devices and human operators, resulting in coordination delays, inconsistent detection performance, and limited throughput under high passenger flow. To address these limitations, this article proposes an edge–cloud collaborative artificial intelligence of things (AIoT) framework that integrates X-ray inspection, robot-side visual perception, safety-score-based fusion, collaborative task allocation model (CTAM)-based task scheduling, and humanoid intervention into a closed-loop perception–decision–execution workflow. For robust carried-item detection in crowded and occluded checkpoint scenarios, an object-aware enhancement (OAE) framework is developed, in which keypoint-guided attention (KGA) uses human pose keypoints to enhance passenger–object interaction features, and a carrying-state classifier (CSC) refines physically inconsistent detections based on human–object carrying relationships. A safety-score-based multimodal fusion strategy further combines X-ray density cues and robot-side visual confidence into a unified safety-screening score for recheck and warning decisions. In addition, a CTAM is formulated to coordinate speech, gesture, locomotion, warning-light, and conveyor-control resources under concurrent security events. Experiments conducted in a controlled real-world metro-checkpoint setting show that the proposed OAE model achieves 46.9% AP, 69.4% AP₅₀, and 60.1% AR. The fusion strategy achieves an area under the curve (AUC) of 0.92. Under the tested controlled metro-checkpoint scenarios, the complete system reduces the average processing time by 46.1% and improves the estimated hourly throughput by 85.5% compared with manual screening.

Index Terms—artificial intelligence of things (AIoT), edge intelligence, humanoid robot, keypoint-guided attention (KGA), metro security screening, multimodal fusion, object-aware detection, task allocation.

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I. INTRODUCTION

IN RECENT years, increasing public-safety risks, terrorist threats, and contraband smuggling have made security checkpoints in public venues [1], [2], [3], [4], such as airports, railway stations, and major event arenas, a critical line of defense for public safety. According to the International Air Transport Association (IATA), global air passenger volume reached 4.8 billion in 2024 and is projected to exceed 7 billion by 2030. This rapidly growing passenger flow poses substantial challenges to conventional security screening systems. However, current checkpoint devices often remain insufficiently connected for smart-city security applications. X-ray machines, metal-detection gates, and manual inspection procedures usually operate as isolated units, leading to coordination latency between digital detection and physical intervention.

Traditional screening workflows rely heavily on human operators to interpret X-ray images and conduct manual inspections. Under high passenger-flow conditions, this model suffers from operator fatigue, subjective judgment, inconsistent detection performance, and high staffing costs. Human-factor studies indicate that vigilance in repetitive visual-search tasks can decline significantly after 20–30 min, a phenomenon known as vigilance decrement [5], [6]. These limitations make manual screening a bottleneck for both security reliability and passenger-processing efficiency.

AI-based X-ray recognition has been widely studied to improve prohibited-item detection. Large-scale benchmarks, such as SIXray [7] and PIDray [8], have promoted the development of attention-based networks [9], Transformer-based detectors [10], and CNN-based or multiscale detection methods for cluttered and overlapping baggage scenarios [11], [12], [13], [14], [15], [16], [17], [18]. Although these methods have advanced image-level recognition, most remain standalone software solutions. They detect prohibited items but do not close the loop from detection results to passenger guidance, conveyor control, warning-device activation, or physical recheck actions. As a result, X-ray machines, metal detectors, and security staff still operate in information silos, which may cause handover delay and potential security risks.

Security robots have also been investigated for patrol, surveillance, hazardous-material detection, and explosive ordnance disposal [19], [20], [21], [22], [23]. Existing systems have demonstrated capabilities in autonomous patrol, multi-

modal sensing, and EOD operations [24], [25], [26], [27]. Mobile inspection robots, such as the system proposed by Saufi et al. [28], further show the feasibility of robotic assistance in security-related scenarios. Nevertheless, these systems are mainly designed for monitoring, patrol, or remote-handling tasks rather than cooperative, high-throughput checkpoint screening. They generally lack real-time integration with X-ray machines, metal-detection gates, conveyor belts, warning devices, and passenger-flow management. Therefore, they cannot directly interpret heterogeneous screening events or generate coordinated robot-executable guidance and intervention actions.

Beyond integrating existing sensing and robotic components, the novelty of the proposed framework lies in its screening-event-driven embodied artificial intelligence of things (AIoT) mechanism. First, the framework establishes a perception-to-action binding mechanism that converts heterogeneous security observations, including X-ray density cues, robot-side visual confidence, and passenger-object carrying states, into robot-executable recheck, warning, and guidance actions. Second, the proposed object-aware enhancement (OAE) module introduces passenger-object relationship modeling through keypoint-guided attention (KGA) and a carrying-state classifier (CSC), enabling the system to determine whether an object is physically associated with a passenger rather than treating objects as isolated detection targets. Third, a collaborative task allocation model (CTAM) extends task scheduling from digital computation and communication resources to embodied interaction resources, including speech, gestures, locomotion, warning lights, and conveyor control. Therefore, the proposed framework forms a closed-loop embodied AIoT control mechanism that links security perception, safety-score interpretation, task allocation, and physical intervention in metro checkpoint scenarios.

To bridge the gap between digital sensing and physical intervention, we propose an AIoT-enabled collaborative security screening system. The main contributions of this article are summarized as follows.

- 1) An OAE detection framework is developed for carried-item detection in metro security screening. KGA uses human pose keypoints as spatial anchors to enhance passenger-object interaction features, while CSC models carry relationships and suppress physically inconsistent detections under occlusion and crowded passenger flow.
- 2) A safety-score-based multimodal fusion method is proposed to integrate X-ray density cues and robot-side visual confidence. The heterogeneous sensing outputs are converted into a unified safety-screening score, where a higher score indicates a higher level of security concern and directly supports recheck or warning decisions.
- 3) A CTAM is formulated for humanoid-robot-based security intervention. By modeling task priority, exclusive resource locking, blocking time, and emergency preemption, CTAM coordinates speech, gesture, locomotion, warning-light, and conveyor-control resources under concurrent security events.

- 4) A complete closed-loop AIoT security screening system is implemented and evaluated in a metro checkpoint environment. Experimental results demonstrate improved detection performance, reduced end-to-end latency, and higher passenger-processing efficiency compared with the manual baseline under the tested protocol.

The remainder of this article is organized as follows. Sections II and III review related work and present the system architecture, respectively. Section IV details the proposed modules. Section V reports the experimental results and Section VI concludes the article.

II. RELATED WORKS

A. AIoT Resource Management and Edge-Cloud Orchestration

Low-latency orchestration is essential for real-time AIoT systems, where sensing, communication, computation, and actuation are tightly coupled. Recent studies on URLLC service function chains and multitier cloud-edge task offloading have investigated latency differentials, backup deployment, transmission delay, processing delay, and resource allocation for latency-sensitive services [29], [30]. Edge-computing-powered AIoT and fog-embedded deep learning further highlight the role of end-edge-cloud collaboration and edge intelligence in reducing cloud dependence and improving real-time responsiveness [31], [32]. In parallel, IoT and AIoT studies have addressed data collection, distributed storage, energy optimization, resource allocation, and task distribution using federated learning, periodic scheduling, decision trees, support vector machines, LSTM-based prediction, multicriteria decision making, random forest, clustering, and reinforcement learning [33], [34], [35], [36], [37]. These studies provide useful insights into AIoT resource management, but they mainly focus on data, storage, energy, computation, or network resources rather than embodied physical intervention.

Edge-cloud robotic platforms and Internet of Robotic Things (IoRT) systems extend AIoT from passive sensing to robotic perception and actuation. Cloud robotics has explored knowledge sharing, computation offloading, and scalable robotic service deployment, while recent IoRT, industrial robotics, and mobile robotic studies have investigated integrated sensing, computing, communication, adaptive sensing, AGV scheduling, and digital-twin-assisted robot navigation [38], [39], [40]. However, these works mainly target general robotic workflows, navigation, adaptive sensing, or industrial logistics. They rarely consider security-checkpoint-specific coordination among X-ray machines, metal-detection gates, warning lights, conveyor belts, passenger states, and humanoid robot actions. In contrast, metro security screening requires the joint orchestration of digital sensing results and embodied physical resources. The proposed CTAM, therefore, extends AIoT resource orchestration from computation and communication tasks to robot-executable intervention resources, including speech, gestures, locomotion, warning lights, and conveyor control. This design links security-event perception, safety-score interpretation, task allocation, and physical

intervention into a closed-loop embodied AIoT workflow for high-throughput metro checkpoint collaboration.

B. Multimodal Semantic Fusion and Reasoning

Multimodal semantic fusion has been widely studied in vision-language understanding, visual question answering, and cross-modal retrieval. For example, KTMN introduces a knowledge-driven two-stage modulation mechanism for visual question answering, showing that task-related knowledge can guide cross-modal reasoning more effectively than simple feature concatenation [41]. Multilevel semantic consistency alignment further improves image-text matching by enforcing semantic consistency across representation levels [42]. These studies suggest that heterogeneous information should be aligned according to task objectives rather than directly combined at the feature or score level.

This principle is also relevant to smart security screening, where X-ray images, robot-side visual observations, metal-detection signals, and passenger-interaction states describe different aspects of the same security event. AI-based prohibited-item detection has been widely studied on benchmarks, such as SIXray and PIDray [7], [8]. CNN-based detectors, attention mechanisms, multiscale aggregation, Transformer-based models, dual-view analysis, and frequency-aware representations have improved detection under overlap and visual clutter [9], [10], [12], [14], [16]. Nevertheless, these methods mainly operate at the image-recognition level, and their outputs are rarely converted into robot-executable decisions. In contrast, the proposed safety-score fusion strategy aligns X-ray density cues and robot-side visual confidence into a unified safety-screening score, enabling direct coordination with CTAM-based humanoid intervention.

C. Robust Perception and Localization

Robust perception and localization are critical for intelligent systems operating in complex public environments. Recent pose-optimization studies improve fine-grained cross-view localization through robust iterative pose refinement, highlighting the importance of spatial consistency under viewpoint variation and localization uncertainty [43]. Cross-temporal progressive enhancement methods further improve object-level detection and weak-feature recovery in challenging visual scenes [44]. Although these studies target different domains, they provide useful insights into robust representation learning, spatial reasoning, and fine-grained perception.

Similar challenges arise in metro security checkpoints, where carried items may be occluded, small objects may be difficult to distinguish, and dense passenger flow may cause unstable human-object associations. Security and inspection robots have been explored for patrol, surveillance, hazardous-material detection, critical-asset inspection, and explosive-ordnance disposal [19], [20], [22], [23]. However, most of these systems focus on monitoring or inspection rather than high-throughput checkpoint collaboration. They usually do not receive real-time X-ray results or structured safety-screening scores, nor do they coordinate with conveyor belts, warning lights, and passenger-flow management. To address

these limitations, the proposed OAE model uses KGA and carrying-state refinement to improve object-aware perception under occlusion and crowded conditions.

Overall, existing studies have advanced individual aspects of X-ray recognition, robotic inspection, semantic fusion, latency optimization, and robust perception. In contrast, the proposed framework couples multisource security sensing, OAE-based object-aware perception, safety-score fusion, and CTAM-driven humanoid intervention within a unified edge-cloud collaborative AIoT architecture for metro checkpoint scenarios.

III. SYSTEM ARCHITECTURE DESIGN

A. System Design Goals

The proposed system is designed according to four criteria: efficiency, accuracy, reliability, and user-friendliness. First, it aims to reduce passenger processing time and increase checkpoint throughput by optimizing task scheduling and minimizing unnecessary robot movements. Second, it seeks to improve contraband detection with high precision and recall while reducing false alarms under complex screening conditions. Third, the system is expected to support continuous operation with fault-tolerance mechanisms, such as self-diagnosis and recovery from sensor or network anomalies. Finally, it provides clear multimodal interaction and socially compliant guidance to reduce passenger anxiety and cognitive load in public metro environments.

B. Overall System Architecture

To enable interoperability between legacy security equipment and robotic controllers, the proposed system adopts a heterogeneous network architecture with the humanoid robot as the primary edge node. EtherCAT is used as the real-time control bus for joint-level actuation, providing deterministic command delivery with a cycle time of <1 ms. MQTT is adopted as the event message bus for high-level data exchange between perception and decision modules, enabling scalable fusion of X-ray images, metal-detection signals, and robot-side sensing data without congesting the control bus.

As shown in Fig. 1, the system is organized into four layers: hardware, perception, decision, and interaction.

- 1) *Hardware Layer*: The hardware platform is a 40-DOF humanoid robot with EtherCAT-based joint communication, an onboard computer equipped with an Intel i9-13900 CPU, 64 GB RAM, and 500 GB SSD, and a 60-V, 12-Ah hot-swappable battery pack for continuous operation.
- 2) *Perception Layer*: This layer acquires multimodal information from passengers, objects, and security devices. A 1080P stereo camera, YOLOv8-Pose, and DeepSORT are used for passenger tracking and behavior analysis, while X-ray images, metal-detector signals, and warning-light states are collected through Ethernet and serial interfaces for security-event perception.
- 3) *Decision Layer*: This layer performs result fusion, workflow control, and anomaly response. The AI analysis

TABLE I
SECURITY EQUIPMENT INTEGRATION

Equipment	Interface	Function
X-ray machine	Ethernet/TCP-IP	Bag-ID matching; image transfer < 200 ms
Conveyor belt	Relay signal	Start/stop control for abnormal-item positioning
Metal gate	RS-485	Zone/intensity parsing and passenger association
Warning lights	RS-232/Serial	CTAM-triggered warning within 100 ms

module integrates X-ray data, metal-detection signals, and robot-side visual observations to generate a safety-screening score. If the score exceeds a preset threshold, CTAM triggers the corresponding recheck, warning, or intervention task.

- 4) *Interaction Layer*: This layer provides passenger-facing guidance through speech, gestures, and warning lights. The voice module integrates ASR, NLU, and TTS functions for natural language interaction, while the visual module uses warning-light states to indicate normal, abnormal, or emergency screening conditions.

To support the layered architecture, the humanoid robot platform integrates a 40-DOF body, EtherCAT-based joint communication, an Intel i9-13900 onboard computer, a 1080P stereo camera, LiDAR, IMU, a 6-MIC array, and a 60-V/12-Ah hot-swappable battery. These components provide the physical basis for perception, decision-making, multimodal interaction, and embodied task execution. The interfaces and synchronization mechanisms of the integrated security equipment are summarized in Table I.

IV. FUNCTIONAL MODULE IMPLEMENTATION

The proposed system comprises four collaborative modules: object-aware vision inspection, multisource data fusion, CTAM-based task allocation, and multimodal interaction.

A. OAE Vision Inspection Module

To address the challenges of high passenger density and frequent occlusions in metro security scenarios, we propose an OAE detection framework. The design of OAE is based on the principle that carried-item detection should be guided not only by visual appearance, but also by human-object spatial relations. Inspired by recent AIoT and IoRT studies that emphasize the joint coordination of sensing, computation, and task execution [31], OAE introduces human keypoints as structural anchors for object localization in metro checkpoint scenes. Unlike generic detectors that treat objects as isolated targets, OAE explicitly models the contextual association between passengers and their carried items. Specifically, KGA enhances image regions that are spatially related to human body parts, while the CSC verifies whether the detected object is physically consistent with the passenger's carrying state. This design enables the detector to reason about passenger-object relationships under occlusion and crowded

passenger flow. As shown in Fig. 2, this module comprises three key components.

1) *Parallel Perception Backbone*: We construct a dual-stream network to jointly model visual appearance and human structural information, enabling robust perception under occlusion and complex backgrounds.

The visual stream employs CSPDarknet53 to extract multi-scale visual features from RGB images. Given an input image $I \in \mathbb{R}^{H \times W \times 3}$, the backbone generates hierarchical feature representations

$$f_v = \{f_v^3, f_v^4, f_v^5\} \quad (1)$$

where shallow features preserve texture and edge details, while deeper features capture high-level semantic context. Multiscale representations improve the detection of small, occluded, and scale-varying objects.

In parallel, the Pose Stream adopts a frozen HRNet-W32 to estimate 17 human body keypoints

$$\mathcal{K} = \{k_1, \dots, k_{17}\}, \quad k = (x, y, c) \quad (2)$$

where each keypoint contains the spatial coordinates and confidence score of a body joint, including shoulders, elbows, wrists, hips, knees, and ankles. By mapping the 17 keypoints onto the image, we generate the keypoint feature $f_k^{(n)}$. These keypoints provide structural cues about human posture and interaction regions.

2) *Keypoint-Guided Attention*: To explicitly model the correlation between body parts and carried items, we compute the attention weights $\alpha_{i,j,n}$. Let $f_v^{(i,j)}$ be the visual feature at spatial location (i, j) , and $f_k^{(n)}$ be the feature of the n th body keypoint. The raw correlation score $e_{i,j,n}$ is computed via a dot-product interaction

$$e_{i,j,n} = \frac{(W_q f_v^{(i,j)})^T (W_k f_k^{(n)})}{\sqrt{d_k}} \quad (3)$$

where W_q and W_k are learnable projection matrices, and d_k is the dimension scaling factor. The final attention weight is obtained using a softmax operation along the keypoint dimension

$$\alpha_{i,j,n} = \frac{\exp(e_{i,j,n})}{\sum_{m=1}^{17} \exp(e_{i,j,m})} \quad (4)$$

where $\alpha_{i,j,n}$ indicates the relevance of keypoint k_n to location (i, j) . The visual features are then refined as follows:

$$\tilde{f}_v[i, j] = f_v[i, j] + \beta \sum_{n=1}^{17} \alpha_{i,j,n} \cdot J_n \quad (5)$$

where β is a hyperparameter balancing original and enhanced features. J_n extracts keypoint-centric features by pooling regions around each keypoint.

3) *Carrying-State Refinement*: The fused feature $\tilde{f}_v$ is fed into the detection head to obtain an initial detection result $\mathbf{D} = \{\text{bbox}_i, \text{cls}_i, \text{conf}_i\}$. Meanwhile, $\tilde{F}_v$ is also input into the CSC, which predicts the human-object interaction state (e.g., hand-held and backpacked) to suppress false positives that are physically inconsistent with human poses. Then, according to

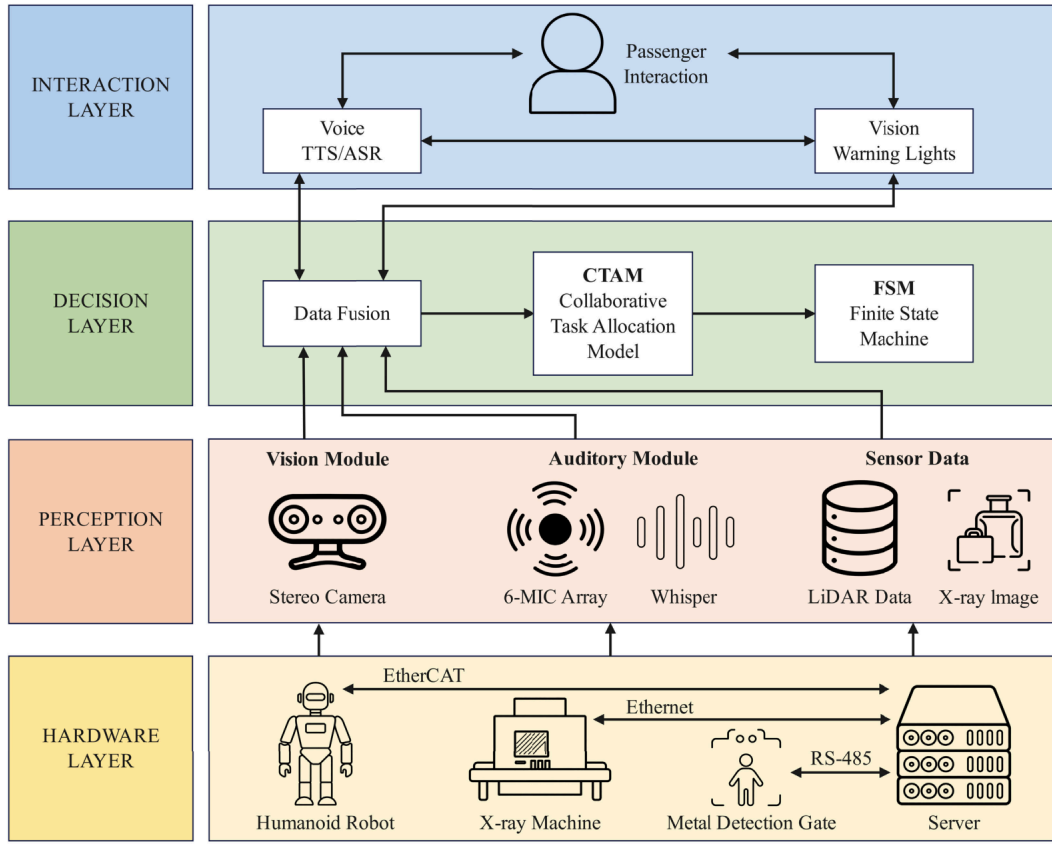


Fig. 1. Overall system architecture.

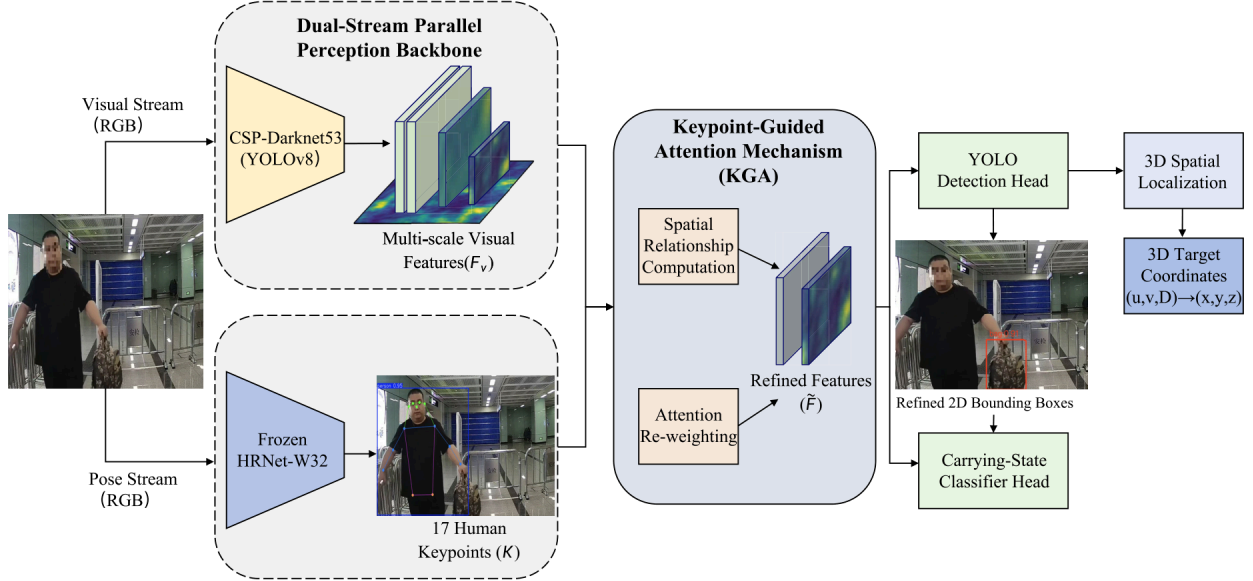


Fig. 2. OAE detection framework.

the predicted carrying state, we further refine $\mathbf{D}$ using state-specific heuristic rules. For example, under the hand-held state, the detection box is slightly adjusted toward the hand region.

To improve reproducibility, the CSC branch takes the region-of-interest (RoI) features of the detected object, the associated passenger region, and their geometric descriptor as input, and outputs the carrying state among hand-held, back/shoulder-carried, dragged, and noncarried states. For

evaluation, the first three states are merged into the carried state, while the last state is treated as noncarried. The OAE module is trained with a multitask objective

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \lambda_{\text{CSC}} \mathcal{L}_{\text{CSC}} \quad (6)$$

where $\mathcal{L}_{\text{det}}$ denotes the detection loss, $\mathcal{L}_{\text{CSC}}$ denotes the carrying-state classification loss, and λ_{CSC} is a weighting coefficient used to balance the two objectives.

B. X-Ray Data Fusion and Safety-Score Assessment

This module receives X-ray images through Ethernet and fuses them with robot-side visual perception results. To support security risk assessment and recheck decision-making, we define a quantitative Safety Score. In this work, the Safety Score denotes a safety-screening score rather than a “safe-level” score; a larger value indicates a higher level of security concern.

Let $\bar{D}$ denote the average X-ray density of the inspected item, $N(\bar{D}) \in [0, 1]$ denote its normalized value, and $s \in [0, 1]$ denote the confidence score from the robot-side visual detector. The safety score is defined as

$$S_{\text{safe}} = \alpha N(\bar{D}) + (1 - \alpha)s \quad (7)$$

where $\alpha \in [0, 1]$ is the fusion weight calibrated on the validation set and can be adjusted according to the current security alert level.

The final recheck decision is determined by a threshold τ

$$y_{\text{recheck}} = \begin{cases} 1, & S_{\text{safe}} \geq \tau \\ 0, & S_{\text{safe}} < \tau \end{cases} \quad (8)$$

where $y_{\text{recheck}} = 1$ indicates that a recheck or warning action should be triggered, and $y_{\text{recheck}} = 0$ indicates normal passage. The generated decision is then sent to CTAM for downstream task scheduling and robot-executable intervention.

The linear form in (7) is adopted to provide an interpretable safety-screening decision. In this formulation, the X-ray density term $N(\bar{D})$ and the robot-side visual confidence term s contribute explicitly to the final score, allowing the source of a recheck decision to be traced at the modality level. Since both terms are weighted by nonnegative coefficients, S_{safe} is monotonically nondecreasing with respect to each modality. Therefore, stronger security evidence from either the X-ray branch or the visual branch will not reduce the final safety-screening score. This property makes the fusion result transparent and suitable for safety-critical metro security screening.

C. Collaborative Task Allocation Model

To coordinate multimodal interaction and robot-control resources under concurrent security events, CTAM extends AIoT resource scheduling from digital resources to embodied intervention resources. Existing AIoT resource-management methods mainly optimize data, computation, storage, energy, or network resources [33], [34], [35], [36], [37], while edge-cloud robotic platforms mainly address computation offloading, service deployment, navigation, or industrial workflow scheduling [32], [38], [39]. In contrast, metro security intervention requires mutually exclusive control over speech, gesture, locomotion, warning lights, and conveyor operation.

Therefore, CTAM formulates these physical actions as schedulable resources and models their priority, locking, blocking, and emergency preemption mechanisms. Specifically, CTAM is implemented as a preemptive fixed-priority scheduler with exclusive resource locking. Unlike the finite-state machine (FSM), which only defines the nominal

workflow, CTAM explicitly handles resource contention and emergency interruption at the embodied edge node.

1) *Task and Resource Modeling*: Let $\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_m\}$ denote the set of event-driven tasks. Each task is represented as

$$\tau_i = \langle a_i, C_i, D_i, P_i, \mathcal{R}_i, T_i \rangle \quad (9)$$

where a_i is the arrival time, C_i is the worst case execution time, D_i is the relative deadline, P_i is the priority level, $\mathcal{R}_i$ is the set of required exclusive resources, and T_i is the minimum interarrival time. A larger P_i indicates a higher priority.

The shared resource set is defined as

$$\mathcal{R} = \{r_{\text{voice}}, r_{\text{gesture}}, r_{\text{locomotion}}, r_{\text{light}}, r_{\text{conveyor}}\}. \quad (10)$$

A resource conflict occurs when two tasks require at least one common resource

$$\mathcal{R}_i \cap \mathcal{R}_j \neq \emptyset. \quad (11)$$

When a higher-priority task arrives, CTAM preempts the lower-priority task and releases or aborts the corresponding resource locks according to the safety policy. Tasks with the same priority are executed according to the first-come, first-served rule.

2) *Scheduling Latency Analysis*: For task τ_i , let $\text{hp}(\tau_i) = \{\tau_k \mid P_k > P_i\}$ denote the set of higher-priority tasks. The worst case response time of τ_i is denoted by W_i and is iteratively computed as

$$W_i^{(q+1)} = C_i + B_i + \delta_{\text{sw}} + \sum_{\tau_k \in \text{hp}(\tau_i)} \left\lceil \frac{W_i^{(q)}}{T_k} \right\rceil C_k \quad (12)$$

where $W_i^{(0)} = C_i$, δ_{sw} is the context-switching overhead, and B_i is the maximum blocking time caused by lower-priority tasks. In our implementation, $\delta_{\text{sw}} < 10$ ms on the onboard Intel i9-13900 platform.

The blocking time is calculated as

$$B_i = \max_{\substack{P_j < P_i \\ \mathcal{R}_j \cap \mathcal{R}_i \neq \emptyset}} C_j^{\text{cs}} \quad (13)$$

where C_j^{cs} is the maximum nonpreemptive critical-section duration of a lower-priority task τ_j that shares resources with τ_i . A task is schedulable if

$$W_i \leq D_i. \quad (14)$$

3) *Emergency Preemption Guarantee*: For the highest-priority emergency task τ_{emg} , CTAM assigns the maximum priority $P_{\text{emg}} = 6$ and locks all physical resources, i.e., $\mathcal{R}_{\text{emg}} = \mathcal{R}$. Since no higher-priority task exists, the emergency task has no higher-priority interference. In addition, CTAM applies hard preemption to emergency events, so any lower-priority task occupying emergency-related resources is immediately aborted, and its locks are released. Thus, $B_{\text{emg}} = 0$, and the worst case response time of the emergency task is bounded by

$$W_{\text{emg}} = C_{\text{emg}} + \delta_{\text{sw}}. \quad (15)$$

With $C_{\text{emg}} = 400$ ms and $\delta_{\text{sw}} < 10$ ms, we obtain

$$W_{\text{emg}} < 410 \text{ ms} \leq D_{\text{emg}} = 500 \text{ ms} \quad (16)$$

which verifies that the emergency task is schedulable under CTAM.

TABLE II
PRACTICAL CTAM RESOURCE-CONFLICT HANDLING POLICY

Resource	Typical Conflict	CTAM Policy	Recovery Strategy
Speech	Two prompts overlap	Higher-priority prompt interrupts the lower-priority one.	Low-priority prompt is discarded if outdated or rescheduled if still relevant.
Gesture	Guidance gesture conflicts with emergency gesture	Emergency gesture preempts; ordinary gesture is stopped after returning to a safe posture.	The robot returns to a neutral pose before executing the next gesture.
Locomotion	Passenger guidance conflicts with re-check positioning	Locomotion is locked during the active motion segment; higher-priority motion waits for the short critical section to finish.	Interrupted guidance is either resumed from the current position or cancelled according to event status.
Warning light	Normal-pass and alarm states overlap	Alarm state overrides normal state immediately.	Light state is reset after staff confirmation or event clearance.
Conveyor control	Normal flow conflicts with abnormal-bag handling	Conveyor stop has the highest priority and is executed immediately.	Conveyor resumes only after manual confirmation or clearance from the CTAM state machine.

4) *Practical Implementation Details*: CTAM is implemented as an event-driven resource manager at the robot edge node. Each incoming security event is converted into a task descriptor containing its priority, deadline, required resources, and recovery policy. The task is inserted into a priority queue and dispatched only after CTAM verifies that the required resources are available. If a resource conflict occurs, a higher-priority task preempts the lower-priority one, whereas tasks with the same priority follow the first-come, first-served rule.

Table II summarizes the practical conflict-handling policies. Speech and warning-light resources can be preempted because outdated prompts or light states may mislead passengers. Gesture and locomotion resources are protected by short nonpreemptive critical sections to avoid unsafe motion. Conveyor control is treated as a safety-critical resource: once an abnormal item or emergency event is detected, the conveyor-stop command receives the highest priority.

Interrupted tasks are handled according to their safety relevance. Expired notification tasks are discarded, recoverable guidance tasks are reinserted into the queue after the emergency task is completed, and safety-critical tasks remain active until the abnormal event is cleared. For conveyor synchronization, CTAM first locks the conveyor resource and issues the stop command; after acknowledgment, it activates the warning light, speech prompt, and corresponding gesture. If the acknowledgment times out, CTAM triggers fallback actions, including retransmission, red-light warning, and staff notification.

D. Multimodal Interaction and Execution

The interaction layer executes the action commands dispatched by CTAM through coordinated speech, gesture, warning-light, and optional conveyor-control outputs. Its purpose is not only to inform passengers of the current screening result, but also to provide clear physical guidance for recheck, waiting, or emergency intervention.

1) *Event-to-Action Mapping*: A deterministic event-to-action mapping is defined in Table III. Each security event is associated with a priority level, a natural-language prompt, and a corresponding interaction primitive. For example, a

liquid-detection event triggers a voice prompt instructing the passenger to remove the liquid item, while the robot simultaneously performs a pointing gesture toward the recheck area. For high-risk events, CTAM dispatches higher-priority warning actions, such as red warning-light activation, checkpoint pause, and emergency speech notification.

2) *Multimodal Output Synchronization*: To improve intelligibility in noisy metro environments, the speech, gesture, and warning-light outputs are synchronized after CTAM dispatches an action command. Specifically, the onset difference among speech output, warning-light activation, and gesture initiation is constrained within 100 ms. This 100-ms threshold refers only to the coactivation synchronization of multimodal outputs after command dispatch, rather than the full end-to-end sensing-to-actuation latency.

E. End-to-End System Latency Model

The latency model is designed from a system-level AIoT perspective rather than from a single-module inference perspective. Edge-AIoT and fog-intelligence studies show that real-time performance depends on the coupled delay of sensing, communication, computation, and actuation [31], [32]. Similarly, IoRT and edge-cloud robotic platforms require coordinated timing across perception, decision-making, and service execution. In the proposed system, the CTAM analysis characterizes only the scheduling latency after an event enters the task allocator, whereas the complete response further involves sensing, X-ray transfer, preprocessing, edge inference, safety-score fusion, MQTT communication, command transmission, and physical actuation. Based on this system-level perspective, we define an event-dependent end-to-end latency model. Specifically, the checkpoint response is decomposed into three practical paths: vision-triggered, X-ray-triggered, and fusion-triggered responses. This event-source-aware formulation prevents time-critical actions from unnecessarily waiting for the slowest sensing branch, such as X-ray image transfer, and provides a clearer basis for evaluating the full sensing-decision-actuation latency.

TABLE III
EVENT-TO-ACTION MAPPING USED BY THE HRI MODULE

Event key	Suggested speech prompt	Priority	Gesture
Carry bag	Please remove your backpack and go through security.	1	Yes
Carry liquid	Liquid detected. Please remove it and place it in the liquid re-check area as instructed.	2	Yes
Miss detection	Possible missed screening. Please wait behind the yellow line and follow the re-screening instructions or wait for staff assistance.	3	Yes
Liquid recheck	There is liquid inside the bag; please cooperate by opening and inspecting it.	4	Yes
Bag abnormal	Abnormality detected in the bag. Please pause and cooperate with the re-inspection. Thank you for your cooperation.	5	Yes
Liquid abnormal	The liquid detection is abnormal. Please retest the liquid.	5	Yes
Emergency	A significant risk of prohibited items has been detected. The security checkpoint is temporarily closed. Please remain where you are and wait. Emergency personnel have taken over the situation.	6	Yes

For the k th security event, the end-to-end physical response latency is defined as

$$L_{e2e}^{(k)} = t_{act_onset}^{(k)} - t_{trigger}^{(k)} \quad (17)$$

where $t_{trigger}^{(k)}$ is the timestamp when the event is generated by the sensing layer, and $t_{act_onset}^{(k)}$ is the timestamp when the first observable robot response, warning-light change, or speech output starts.

We further define the decision latency as

$$L_{dec}^{(k)} = t_{dispatch}^{(k)} - t_{trigger}^{(k)} \quad (18)$$

where $t_{dispatch}^{(k)}$ denotes the time when CTAM dispatches the corresponding action command. Thus, the difference between L_{e2e} and L_{dec} mainly reflects command transmission and physical actuation onset.

The robot-side visual branch and X-ray branch are defined as

$$L_{vis} = L_{cam} + L_{pre} + L_{oae} \quad (19)$$

$$L_{xr} = L_{xfer} + L_{xray} \quad (20)$$

where L_{cam} , L_{pre} , and L_{oae} denote camera acquisition, preprocessing, and OAE inference latency, respectively. L_{xfer} and L_{xray} denote X-ray image transfer and X-ray risk-estimation latency, respectively.

After the perception result is obtained, the common communication, scheduling, and actuation latency is

$$L_{post} = L_{mqtt} + L_{sched} + L_{cmd} + L_{act} \quad (21)$$

where L_{mqtt} is the MQTT publish/subscribe latency, L_{sched} is the CTAM scheduling latency, L_{cmd} is the command transmission latency, and L_{act} is the physical actuation onset latency.

According to the event source and decision logic, the end-to-end latency is divided into three practical paths. For vision-triggered events, such as passenger-carried-item detection, the system does not wait for X-ray transfer

$$L_{e2e}^{vis} = L_{vis} + L_{post}. \quad (22)$$

For X-ray-triggered events that do not require visual confirmation, the latency is

$$L_{e2e}^{xr} = L_{xr} + L_{post}. \quad (23)$$

For fusion-triggered events that require both robot-side visual information and X-ray-side risk estimation, the two branches are processed in parallel before safety-score fusion

$$L_{e2e}^{fus} = \max(L_{vis}, L_{xr}) + L_{fuse} + L_{post} \quad (24)$$

where L_{fuse} denotes the safety-score fusion latency. This path represents the most complete sensing–fusion–scheduling–actuation process in the proposed framework.

It should be noted that the 100-ms synchronization threshold discussed in the interaction module refers only to the coactivation delay among speech, gesture, and warning-light outputs after CTAM has dispatched an action command. It is not equivalent to the full sensing-to-actuation latency defined in (17).

V. EXPERIMENTAL RESULTS AND ANALYSIS

To quantitatively evaluate the performance of the proposed system, we conducted comparative experiments in a real-world metro security checkpoint environment. The setup replicated the spatial layout, lighting conditions, and passenger flow protocols of standard operational checkpoints to ensure ecological validity. The physical setup of the experimental platform is illustrated in Fig. 3, which demonstrates the complete operational loop from the OAE-based visual detection and X-ray data fusion to the robot's final behavioral execution. The experiments used the hardware platform specified in Section III, with the Intel i9-13900 computing unit for edge inference and the 1080P stereo camera for data acquisition. The evaluation covers visual detection accuracy, ablation analysis, multimodal fusion validation, end-to-end latency, robustness discussion, and operational efficiency compared with the manual baseline.

A. Visual Detection Precision Analysis

1) *Dataset Description*: We collected 16 000 images from eight metro station checkpoint environments. To avoid data leakage, the dataset was split at the environment and scenario levels into 11 200 training images, 1200 validation images, and 3600 test images. The validation set was used only for hyperparameter calibration, including the fusion weight α and decision threshold τ , while the test set was used exclusively for final evaluation. Following standard

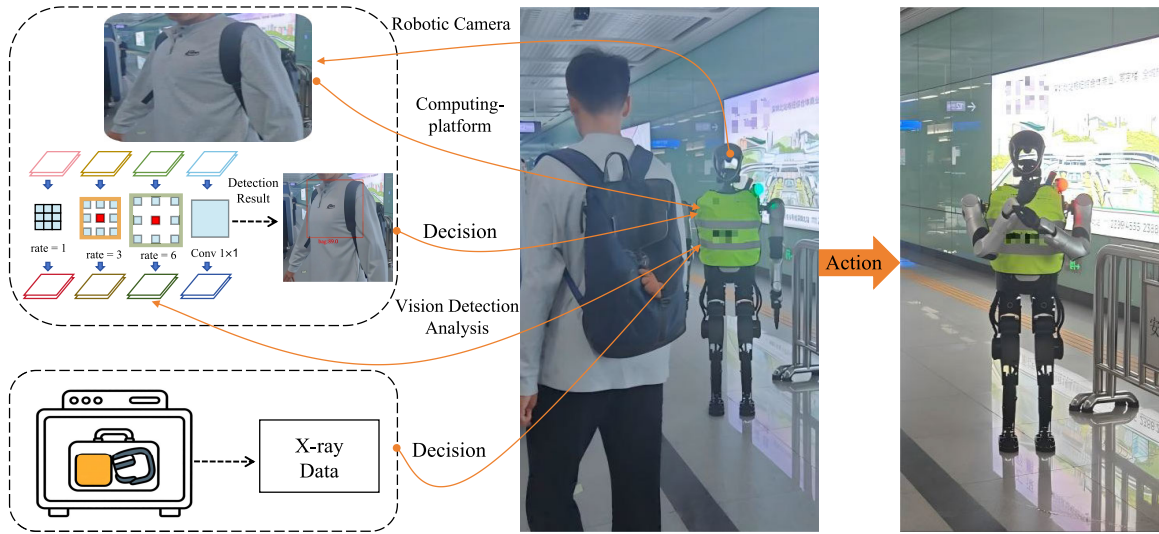


Fig. 3. Overview of the experimental validation platform and operational workflow.

object-detection annotation protocols, three trained annotators used LabImg to annotate object bounding boxes in YOLO format. The basic object categories include bags, suitcases, and bottles, with proportions of 65%, 12%, and 23%, respectively. Each object instance was further assigned a carrying-state label, i.e., carried or noncarried. Therefore, the final evaluation was conducted on six object-state target labels: bag-carried, bag-noncarried, suitcase-carried, suitcase-noncarried, bottle-carried, and bottle-noncarried. To improve annotation consistency, each image was first labeled by one annotator and then cross-checked by a second annotator. Samples with inconsistent bounding boxes, ambiguous categories, or uncertain carrying states were reviewed by the third annotator, and the final labels were determined through consensus. To alleviate class imbalance, stratified sampling was used to preserve object-category and carrying-state proportions across the training, validation, and test sets. During training, balanced sampling and standard data augmentation, including random scaling, cropping, horizontal flipping, brightness variation, and mild blur/noise perturbation, were applied to reduce bias toward dominant bag-related categories. The dataset covers diverse metro checkpoint conditions, including different camera viewpoints, lighting conditions, passenger-flow densities, conveyor-belt layouts, and screening-device positions. It also contains challenging cases with an average occlusion rate of 44.1%, 5.3 individuals per image on average, and 12.3% low-confidence human keypoints, making it suitable for evaluating object detection and passenger-object association under realistic metro security scenarios. For privacy protection, identifiable passenger faces in the visualization examples were anonymized using mosaic masking, and no personally identifiable information was used for model training or evaluation. Due to privacy and metro-operation confidentiality restrictions, the dataset and trained models are not publicly available at this stage. Future release may be considered subject to approval from the corresponding metro authorities and additional privacy review.

 TABLE IV
PERFORMANCE COMPARISON OF VISUAL DETECTION ALGORITHMS

Model	Year	AP	AP ₅₀	AP ₇₅	AR	FPS
YOLOv5-s	2020	30.4	49.1	31.8	41.3	31
YOLOv8-s	2023	36.4	54.4	38.2	46.8	28
RT-DETR-R18 [45]	2023	36.1	58.6	37.5	49.8	24
YOLOv10-s [46]	2024	36.9	55.4	39.1	48.2	29
RT-DETRv2-s [47]	2024	38.3	59.9	40.6	50.7	22
RF-DETR-s [48]	2025	42.1	63.5	44.8	54.6	14
YOLOv12-s [49]	2025	38.8	59.5	41.2	50.3	27
Ours	—	46.9	69.4	49.7	60.1	16

Note: Metrics are based on the COCO protocol. AP denotes mAP@0.5:0.95.

2) *Evaluation Metrics and Protocol*: To ensure standardized comparison, we adopted the MS COCO evaluation protocol. The performance is evaluated on $N = 6$ object-state target labels. The metrics are strictly defined as follows.

- 1) *Average Precision (AP)*: The primary metric averaging precision over ten IoU thresholds (0.50 : 0.05 : 0.95), reflecting comprehensive localization accuracy.
- 2) *AP₅₀ / AP₇₅*: Precision calculated at specific IoU thresholds of 0.50 and 0.75, respectively.
- 3) *Average Recall (AR)*: The maximum recall given a fixed number of detections per image, indicating the model's ability to find objects.
- 4) *FPS*: Frames per second measured on the robot's onboard Intel i9-13900 CPU.

3) *Performance Comparison*: Table IV presents a comprehensive quantitative comparison between our proposed OAE framework and several state-of-the-art real-time detectors, including both recent CNN-based architectures (e.g., YOLOv10-s and YOLOv12-s) and Transformer-based models (e.g., RT-DETRv2-s and RF-DETR-s).

TABLE V
ABLATION STUDY ON PROPOSED MODULE

Configuration	AP	AP ₅₀	AP ₇₅	AR	FPS
YOLOv8-s (Baseline)	36.4	54.4	38.2	46.8	28
+ KGA only	40.6	60.3	42.7	52.4	20
+ CSC only	39.5	59.5	41.5	51.6	27
+ KGA + CSC	43.5	65.1	45.8	56.7	19
+ KGA + CSC + Refinement	46.9	69.4	49.7	60.1	16

The proposed method achieves the highest performance among the compared detectors, with 46.9% AP and 69.4% AP₅₀. Compared to the baseline YOLOv8-s, our method improves the primary AP metric by a substantial + 10.5%. More importantly, when evaluated against the latest attention-centric YOLOv12-s (2025), our framework still demonstrates a significant + 8.1% advantage in AP. Furthermore, compared to the highly competitive RF-DETR-s, our model maintains a + 4.8% lead in AP and a + 4.9% lead in the stringent AP₇₅ metric (49.7% versus 44.8%), indicating stronger bounding-box regression and localization capability and high-precision localization in cluttered backgrounds.

The proposed OAE also improves AR to 60.1%, a 5.5 percentage-point gain over the second-best model, RF-DETR-s. This result indicates improved recall for occluded and densely packed objects, which is important for metro security screening.

Regarding computational efficiency, the integration of the enhancement modules inevitably introduces overhead, resulting in an inference speed of 16 frames per second (FPS). While slower than the lightweight YOLO variants, it operates faster than the RF-DETR-s baseline (14 FPS). Given the substantial accuracy improvement, although OAE introduces additional computation, its 16-FPS inference speed remains sufficient for the tested real-time screening workflow.

B. Ablation Study

To verify the contribution of each component, we conducted a stepwise ablation study as shown in Table V.

1) Impact of Individual Modules:

- 1) *Effect of KGA*: Incorporating the KGA module alone improves the AP₅₀ from 54.4% to 60.3% (+5.9%). This indicates that leveraging human pose information effectively guides the network to focus on carried items, recovering features that are partially occluded.
- 2) *Effect of CSC*: Adding the CSC independently yields an AP₅₀ of 59.5%. By filtering out physically impossible object-person associations (e.g., floating bags), CSC effectively improves precision.

2) *Synergistic Integration*: When both KGA and CSC are combined, the performance further boosts to an AP₅₀ of 65.1%. The inclusion of the final refinement step (3-D localization and temporal smoothing) brings the final AP₅₀ to 69.4% and the comprehensive AP to 46.9%. This consistent improvement, corresponding to a 10.5 percentage-point AP gain over the baseline, confirms that the proposed modules

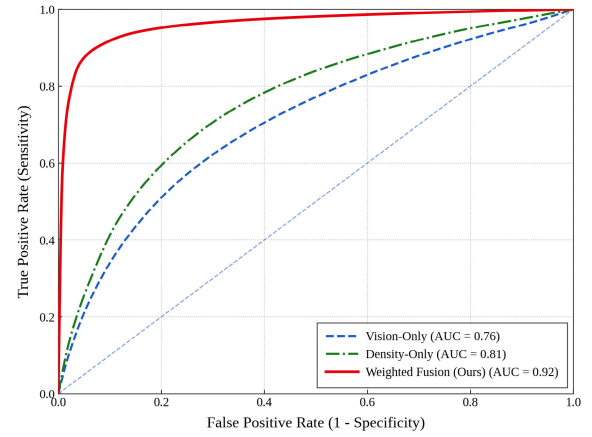


Fig. 4. ROC analysis.

are complementary in a real-world security screening environment.

C. Multimodal Fusion Validation

We calibrated the fusion weight α and decision threshold τ using the held-out validation set of 1200 images, and then evaluated the fusion performance using receiver operating characteristic (ROC) and PR analyses.

1) *Hyperparameter Calibration*: The fusion formula depends on the weighting factor α , which balances the contribution of X-ray density $\mathcal{N}(\bar{D})$ and visual confidence s . We conducted a grid search for $\alpha \in [0, 1]$ with a step size of 0.1.

- 1) *Vision-Dominated* ($\alpha < 0.3$): The system suffers from missed detections of concealed items (e.g., knives inside bags), resulting in a low recall.
- 2) *Density-Dominated* ($\alpha > 0.8$): The system generates excessive false positives due to benign high-density objects (e.g., laptops and metal water bottles).
- 3) *Optimal Balance* ($\alpha = 0.6$): The F1-score peaks at $\alpha = 0.6$, suggesting that density information should slightly outweigh visual probability in high-stakes security scenarios.

Consequently, we fixed $\alpha = 0.6$ for all subsequent experiments.

2) *ROC and PR Analysis*: Figs. 4 and 5 compare the ROC and PR curves of the vision-only, density-only, and proposed fusion strategies.

As illustrated in the ROC analysis, the unimodal methods show limitations: vision-only achieves an area under the ROC curve area under the curve (AUC) of 0.76 due to occlusion, while density-only achieves 0.81 but lacks semantic discrimination. The proposed fusion method improves the AUC to 0.92. Similarly, in the precision-recall (PR) analysis, the fusion model maintains high precision (>0.85) even at high recall levels (>0.9), indicating that the multimodal integration effectively reduces false alarms without compromising safety.

3) *Comparison With Learning-Based Fusion Strategies*: To further evaluate whether a more expressive fusion mechanism can improve decision performance, we compared the proposed linear safety-score fusion with three representative

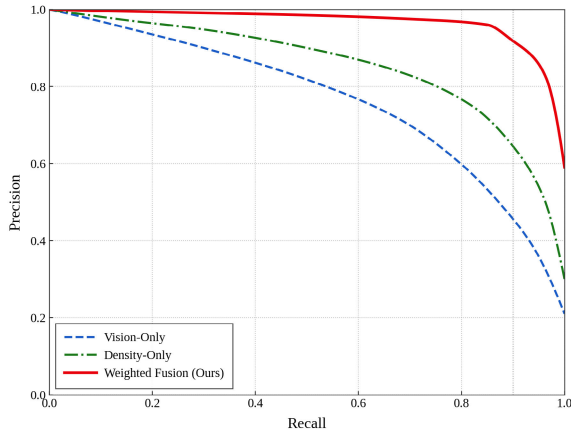


Fig. 5. PR curves.

TABLE VI
COMPARISON WITH LEARNING-BASED FUSION STRATEGIES

Fusion Strategy	AUC	F1	L_{fuse} (ms)	Interp.	Monot.
Logistic Regression	0.915	0.901	1.0	Yes	Yes
MLP Fusion	0.928	0.917	8.4	No	No
Attention Fusion	0.931	0.920	12.1	No	No
Linear (Ours)	0.920	0.908	1.1	Yes	Yes

learning-based alternatives using the same two inputs, namely the normalized X-ray density $N(\bar{D})$ and the robot-side visual confidence s . The compared methods include logistic regression (LR), a two-layer multilayer perceptron (MLP), and a lightweight attention-based fusion module. All methods were trained or calibrated using the same validation set and evaluated under the same operating protocol.

As shown in Table VI, the learning-based fusion strategies achieve slightly higher discriminative performance than the proposed linear fusion. The best result is obtained by attention fusion, which improves the AUC by 1.1 percentage points compared with the proposed method. However, this improvement comes with substantially higher latency, increasing L_{fuse} from 1.1 to 12.1 ms. More importantly, the MLP and attention-based models do not inherently provide per-modality interpretability or monotonicity guarantees unless additional constraints are imposed during training. LR preserves interpretability and monotonicity when nonnegative coefficients are enforced, but its performance is comparable to that of the proposed linear fusion. These results suggest that, in the tested safety-critical and latency-constrained checkpoint setting, the proposed linear safety-score fusion provides a better balance among accuracy, latency, interpretability, and deployment reliability.

4) *Threshold Selection and Failure Analysis*: The decision threshold τ for triggering a manual recheck was determined by maximizing the F1-score on the validation set. We selected $\tau = 0.75$, which corresponds to a false positive rate (FPR) of 4.2% and a true positive rate (TPR) of 96.5%.

Despite the performance gains, we identified two primary failure modes.

TABLE VII
COMPONENT-LEVEL LATENCY BREAKDOWN

Component	Mean (ms)	Max (ms)	Bound (ms)
Camera capture L_{cam}	28.4	33.1	33.3
Preprocessing L_{pre}	2.8	4.8	5.0
OAE inference L_{oae}	55.9	57.3	57.5
X-ray transfer L_{xfer}	156.3	197.8	200.0
X-ray risk L_{xray}	3.6	5.0	5.5
Safety-score fusion L_{fuse}	1.1	1.6	2.0
MQTT L_{mqtt}	6.3	9.8	10.0
CTAM L_{sched}	4.1	8.7	10.0
Command L_{cmd}	0.8	1.2	1.5
Actuation L_{act}	40.4	47.2	50.0

TABLE VIII
END-TO-END LATENCY OF DIFFERENT RESPONSE PATHS

Response Path	Mean (ms)	Max (ms)	Bound (ms)
Vision-triggered $L_{\text{e2e}}^{\text{vis}}$	138.7	162.1	167.3
X-ray-triggered $L_{\text{e2e}}^{\text{xr}}$	211.5	269.7	277.0
Fusion-triggered $L_{\text{e2e}}^{\text{fus}}$	212.6	271.3	279.0

- 1) *High-Density Benign Objects*: Thick metal objects (e.g., camera lenses and thermoses) generate high $N(\bar{D})$ scores, occasionally pushing the weighted sum above τ even when visual confidence is low. This accounts for approximately 65% of the false positives.
- 2) *Low-Density Contraband*: Certain prohibited items made of composite materials (e.g., 3-D-printed plastic weapons) have low X-ray density and, if visually occluded, may result in false negatives. Future work will incorporate dual-energy X-ray analysis to better distinguish material atomic numbers.

D. End-to-End Latency Evaluation

To evaluate the real-time performance of the complete AIoT security loop, we measured the latency from event generation to the first observable robot response. Timestamps were recorded at the camera interface, X-ray receiving module, OAE inference engine, safety-score fusion module, MQTT publisher/subscriber, CTAM scheduler, actuator command interface, and interaction output. External device events were timestamped at the receiving interface of the robot edge node to reduce clock-offset errors. A total of $N = 200$ independent trials were collected under the same experimental settings as the efficiency evaluation.

As shown in Table VII, X-ray transfer and OAE inference are the dominant contributors to the total latency, whereas MQTT communication and CTAM scheduling account for only a small fraction of the response time. Table VIII further reports the end-to-end latency of the vision-triggered, X-ray-triggered, and fusion-triggered paths, whose measured maximum latencies remain below their analytical bounds. The maximum measured latencies of the vision-triggered, X-ray-triggered, and fusion-triggered paths are 162.1, 269.7, and

TABLE IX
ROBUSTNESS ANALYSIS UNDER OPERATIONAL PERTURBATIONS

Perturbation	Stress Setting	Main Results	Robustness Evidence
Occlusion severity	Four occlusion bins $\rho = 1 - A_{\text{visible}}/A_{\text{bbox}}$	AP ₅₀ : 74.6/69.1/59.3/44.8 from light to severe; YOLOv8-s gains: 10.5–21.3 p.p.	KGA and CSC remain effective under degraded appearance cues.
Illumination variation	Intensity scaling: −50%, −30%, +30%, +50%	AP: 41.2 at −50% and 42.6 at +50%; baseline AP: 46.9	Moderate degradation; performance remains above YOLOv8-s clean-input AP of 36.4.
Sensor noise	Gaussian $\sigma = \{5, 10, 20\}$; salt-and-pepper $\eta = \{1\%, 3\%, 5\%\}$	AP: 37.4 at $\sigma = 20$; 39.6 at $\eta = 5\%$	Gradual degradation; CSC suppresses physically inconsistent noisy detections.
Network instability	Packet loss $\ell = \{0, 5, 10, 20\}\%$; jitter $j = \{0, \pm 50, \pm 100, \pm 200\}$ ms	Vision latency: 138.7–215.4 ms; task completion $\geq 96.8\%$; emergency deadline $\geq 99.5\%$	CTAM preserves safety-critical execution through priority scheduling and fallback policies.
Passenger-flow density	Passenger count: [1, 2], [3, 5], [6, 8], and ≥ 9	AP: 51.4–38.9; APT: 19.4–28.5 s; throughput: 126.3 pass/h at $n_p \geq 9$	OAE and CTAM preserve usable performance under dense passenger flow.

Note: p.p. denotes percentage points; APT denotes average processing time.

271.3 ms, respectively, all of which are below their analytical bounds.

It should be emphasized that the 100-ms threshold in the interaction module refers only to the synchronization delay among speech, gesture, and warning-light outputs after CTAM dispatches an action command. It is not the full sensing-to-actuation latency, which additionally includes perception, X-ray transfer, inference, fusion, MQTT communication, scheduling, command transmission, and actuation onset.

E. Discussion of System Robustness

To further evaluate system reliability beyond nominal detection accuracy and latency, we conducted a robustness analysis under five operational perturbations relevant to metro checkpoint deployment: occlusion severity, illumination variation, sensor noise, network instability, and passenger-flow density. All perturbations were applied at evaluation time on the test set defined in Section V-A, and the model was not retrained. Thus, the results reflect deployment-time robustness rather than adaptation. For perception-related perturbations, AP and AP₅₀ were measured on perturbed test images. For network instability, packet loss and latency jitter were injected into the MQTT communication link, and system-level metrics were averaged over $N = 200$ independent trials per condition. The results are summarized in Table IX.

As shown in Table IX, the proposed system exhibits gradual performance degradation under visual perturbations rather than abrupt failure. Under increasing occlusion, AP₅₀ decreases from 74.6% to 44.8%, but the gain over YOLOv8-s increases from 10.5 to 21.3 percentage points. This indicates that the keypoint-guided geometric prior and carrying-state refinement become more beneficial when object appearance is partially unavailable. Under illumination and noise perturbations, the model also maintains usable detection performance, with AP remaining 41.2% under −50% illumination, 42.6% under +50% illumination, 37.4% under Gaussian noise with $\sigma = 20$, and 39.6% under salt-and-pepper noise with $\eta = 5\%$.

For system-level perturbations, CTAM improves robustness by coordinating priority scheduling, emergency preemption, timeout checking, and fallback execution. Under packet loss and latency jitter, the vision-triggered latency increases from

138.7 to 215.4 ms, while the task-completion rate and emergency-deadline satisfaction remain above 96.8% and 99.5%, respectively. These results indicate that communication disturbances mainly increase response latency rather than causing immediate task failure. Meanwhile, under dense passenger flow, AP decreases from 51.4% to 38.9%, and average processing time (APT) increases from 19.4 to 28.5 s; nevertheless, the system still sustains an estimated throughput of 126.3 passengers/h at $n_p \geq 9$, which remains higher than the manual baseline under the tested protocol.

Overall, the robustness analysis provides additional evidence that the proposed framework can tolerate common nonideal metro checkpoint conditions. The perceptual modules contribute robustness under degraded visual inputs, while CTAM supports stable task execution under communication and crowd-density disturbances. These findings strengthen the reliability of the experimental conclusions, although long-term multistation validation remains necessary for verifying large-scale operational robustness.

F. Comparative Analysis of Efficiency and Operational Benefit

1) *Experimental Protocol and Statistical Controls*: To ensure statistical rigor, the comparative experiments were conducted using a randomized block design. For each of the four scenarios (S1–S4), we conducted $N = 50$ independent trials, resulting in a total of 200 trials. About 20 volunteers with different walking speeds and behavioral patterns acted as passengers to introduce realistic variability. The manual baseline was executed by a fixed team of three certified security officers to control for operator-skill differences, and the acoustic background was maintained below 65 dB to reduce environmental interference. Welch’s two-tailed t -test was used to evaluate the statistical significance of the processing-time differences between manual and robotic operations.

2) *Efficiency Results*: Table X reports the APT, standard deviation (SD), 95% confidence interval (CI), and statistical significance for the four scenarios. The proposed robotic system consistently reduces APT across all scenarios, and all differences are statistically significant ($p < 0.001$). The mean SD also decreases from 5.59 s in the manual mode to 1.19 s in

TABLE X
STATISTICAL COMPARISON OF PROCESSING TIME

ID	Scenario	Manual Mode (s)		Robotic System (s)		p-value
		Mean $\pm$ SD	95% CI	Mean $\pm$ SD	95% CI	
S1	Backpack Only	30.17 $\pm$ 4.25	[28.9, 31.4]	16.50 $\pm$ 0.85	[16.2, 16.8]	< 0.001
S2	+ Liquid (Hand)	36.28 $\pm$ 5.10	[34.8, 37.7]	19.20 $\pm$ 1.10	[18.9, 19.5]	< 0.001
S3	+ Liquid (Bag)	45.60 $\pm$ 6.88	[43.6, 47.6]	24.80 $\pm$ 1.45	[24.4, 25.2]	< 0.001
S4	+ Contraband	45.60 $\pm$ 6.12	[43.8, 47.4]	24.50 $\pm$ 1.34	[24.1, 24.9]	< 0.001

Note: P-values calculated using Welch's t-test. The robotic system shows significantly lower variance (σ), indicating superior operational stability.

the robotic system, indicating more repeatable execution under the same screening protocol.

In the most complex contraband scenario (S4), the processing time is reduced from 45.60 ± 6.12 to 24.50 ± 1.34 s, corresponding to a 46.3% reduction. Across all scenarios, the macro-APT decreases from 39.41 to 21.25 s, yielding an overall reduction of 46.1%. Based on $H_{\text{throughput}} = 3600/\text{APT}$, the estimated hourly throughput improves from 91.3 to 169.4 passengers/h, corresponding to an 85.5% improvement under the tested protocol.

These gains are mainly attributed to reduced information handover delay and more consistent task execution. The MQTT-based event bus synchronizes X-ray results with the robot-side decision engine, while CTAM coordinates speech, gesture, warning-light, and conveyor-control resources under concurrent events. These results suggest potential efficiency gains in controlled checkpoint scenarios, although long-term multistation validation is still required before drawing conclusions about deployment-scale capacity improvement.

VI. CONCLUSION

This article presented an edge-cloud collaborative AIoT framework for smart metro security screening. By integrating X-ray inspection, robot-side visual perception, safety-score-based fusion, CTAM-based task scheduling, and humanoid intervention, the proposed system establishes a closed-loop perception-decision-execution workflow. The OAE module improves carried-item detection by combining KGA-based pose guidance with CSC-based carrying-state refinement, while the safety-score fusion module converts heterogeneous sensing outputs into robot-executable recheck and warning decisions. CTAM further coordinates speech, gesture, locomotion, warning-light, and conveyor-control resources through priority scheduling, resource locking, and emergency preemption.

Experiments in controlled metro-checkpoint scenarios show that the proposed OAE model achieves 46.9% AP, 69.4% AP₅₀, and 60.1% AR. The fusion strategy reaches an AUC of 0.92, and all measured end-to-end latencies remain within their analytical bounds. Compared with the manual baseline under the tested protocol, the proposed system reduces the APT by 46.1% and improves the estimated hourly throughput by 85.5%. These results provide preliminary evidence that the proposed framework can improve short-term checkpoint efficiency.

Several limitations remain. The dataset and trained models cannot be publicly released at this stage due to passenger privacy and metro-operation confidentiality requirements. The safety-score fusion strategy may still produce false positives for benign high-density objects and false negatives for low-density contraband. In addition, the operational evaluation was conducted under controlled scenarios. Future work will focus on dual-energy X-ray analysis, adaptive safety-score calibration, larger-scale multistation robustness validation, long-term field deployment, and total-cost-of-ownership assessment.

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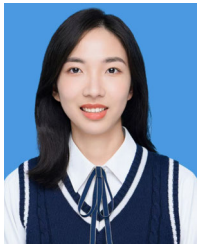
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